DEPARTMENT OF CIVIL ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

B.TECH THESIS PROJECT · 2026

Cradle

Embodied carbon of Indian building materials, with every number cited.

PRESENTED BY

Durgesh Kumar

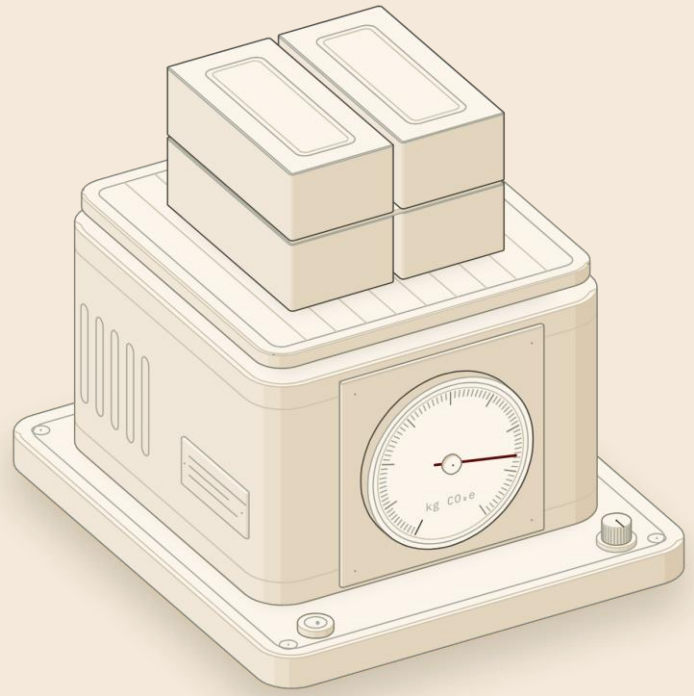
Roll no. 23CE36005

SUPERVISOR

Dr. Vihangraj V. Kulkarni

Assistant Professor, Civil Engineering

Fig 1 CARBON SCALE



1 m³ OF CLAY BRICK WALL 665 kg CO_2e

Half of a building's carbon is in its materials

30%

of India's greenhouse gas emissions come from buildings

48%

of building emissions are embodied: from making the materials, not running the building

SOURCE: NITI AAYOG (2026), SCENARIOS TOWARDS VIKSIT BHARAT AND NET ZERO: BUILDINGS

INDIA HAS STARTED TO ASK FOR THIS NUMBER

ECSBC 2024 — BEE · APPENDIX A6

Report embodied carbon per m^2 for stages A1–A3

GRIHA v2019 — CRITERION 20

Up to 5 points for a lower GWP, using LCA and EPDs

IGBC Net Zero — CARBON RATING

Design limit of 700 kg CO_2e per m^2

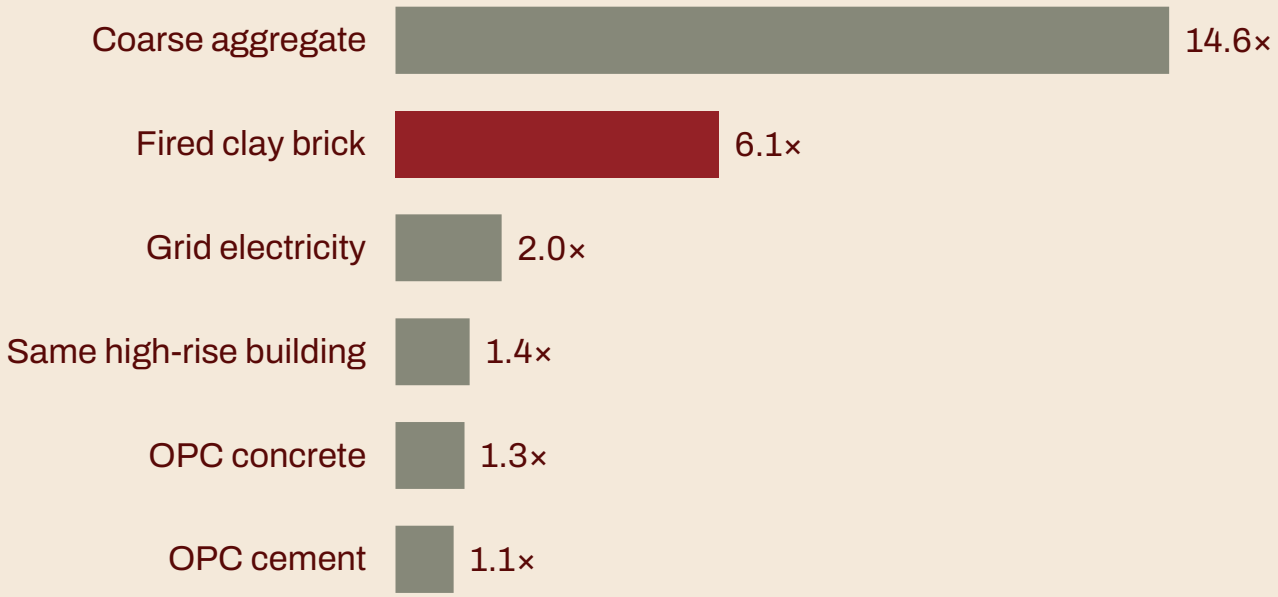
CCTS 2023 — MINISTRY OF POWER

Emission-intensity targets for 186 cement plants

Researchers need a number they can defend.

Published Indian numbers are hard to check

SAME INPUT, DIFFERENT INDIAN SOURCES · HIGHEST ÷ LOWEST VALUE



A fired clay brick is 0.097 to 0.59 kg CO_2e per kg, depending on the source.

2 of 9

Indian studies reproduce when their arithmetic is redone

0 of 9

give a range (uncertainty) on their result

71

South Asian building LCAs reviewed: all used ecoinvent, a paid European database

SOURCES: THESIS FIG. 2.2 AND TABLE 2.2; TALPUR ET AL. (2023)

Aim: a free calculator you can check by hand

01

Factor library

Indian emission factors, each cited to its table or page

02

Engine

EN 15804 modules A1–A3 to D, with an uncertainty range

03

Validation

Reproduce a published Indian study, line by line

04

Application

Compare Indian walls and cements, and test the factor choice

SCOPE · EN 15804 LIFE-CYCLE MODULES

A1–A3	A4	A5	B1–B7	C1–C4	D
Product	Transport to site	Construction	Use stage	End of life	Beyond life
ALWAYS IN	OPTIONAL	OPTIONAL	NOT INCLUDED	OPTIONAL	SHOWN APART

Total = A1–A3 + A4 + A5 + C1–C4 D shown apart

GWP only, kg CO_2e | Building materials | Free data only

Method: the four steps of ISO 14044

STEP 1

Goal and scope

Functional unit
Life-cycle modules
Assumptions

STEP 2

Inventory

Manual entry
BoQ import
AI draft (size only)

STEP 3

Impact

Quantity × factor
For each module
Working on each line

STEP 4

Interpretation

Largest contributors
Monte Carlo range
Indian-data share

STANDARDS

IS/ISO 14040 and 14044
EN 15804+A2

CPWD DAR 2014

Materials per unit of work, from the official rate analysis

Factor library

106 Indian factors, each with source, page and licence

Outputs

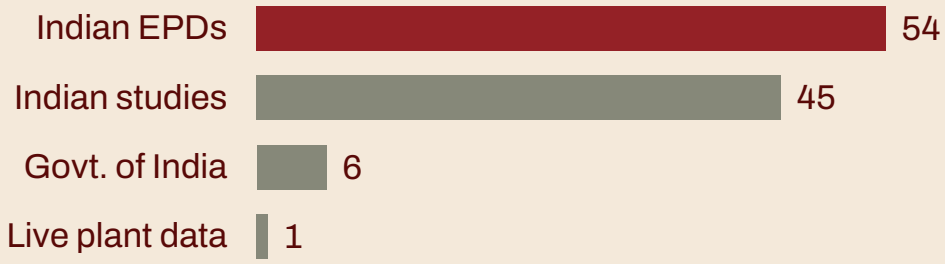
Report, Excel with live formulas, BibTeX, project file

106 Indian factors, each traced to a page

106

emission factors (A1–A3), all for products made in India

WHERE THEY COME FROM



- Free data only
- Indian first
- Read from the document
- Cited to a page

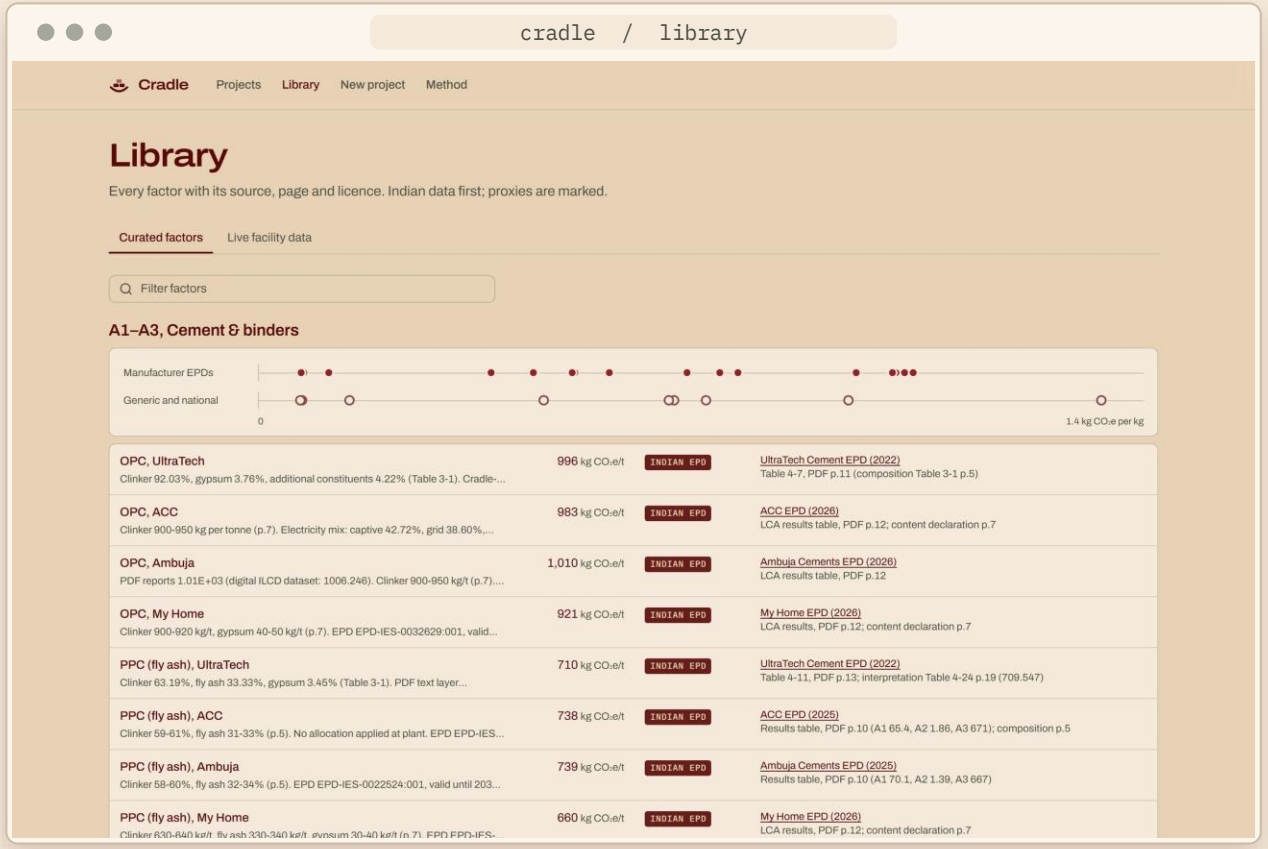


FIG 2 · EACH FACTOR SHOWS ITS EPD, PAGE AND DATA TAG

Every number is a quantity times a factor

A1–A3	Q × EF	
A4	m × d × EF per t·km	RAIL, SHIP
	(m ÷ payload) × d × EF per vehicle·km	TRUCK
A5	waste % × (A1–A3 + A4) + site energy × EF	
C1–C4	m × EF of the end-of-life route	
D	reported beside the total, never added	

Total = A1–A3 + A4 + A5 + C1–C4

Q quantity · EF factor · m mass (t) · d distance (km)

WORKING SHOWN ON EVERY LINE

1,540 kg × 0.39
= 600.5 kg CO_2e

Clay bricks in 1 m^3 of wall, IFC 2017 India average

UNCERTAINTY

Monte Carlo, 5,000 runs

Triangular draws on each published range; 95% band; seed 42

DATA STRENGTH

Indian-data share

Each kg tagged by its source, e.g. 83% Indian, 17% proxy

Describe the work; CPWD supplies the materials

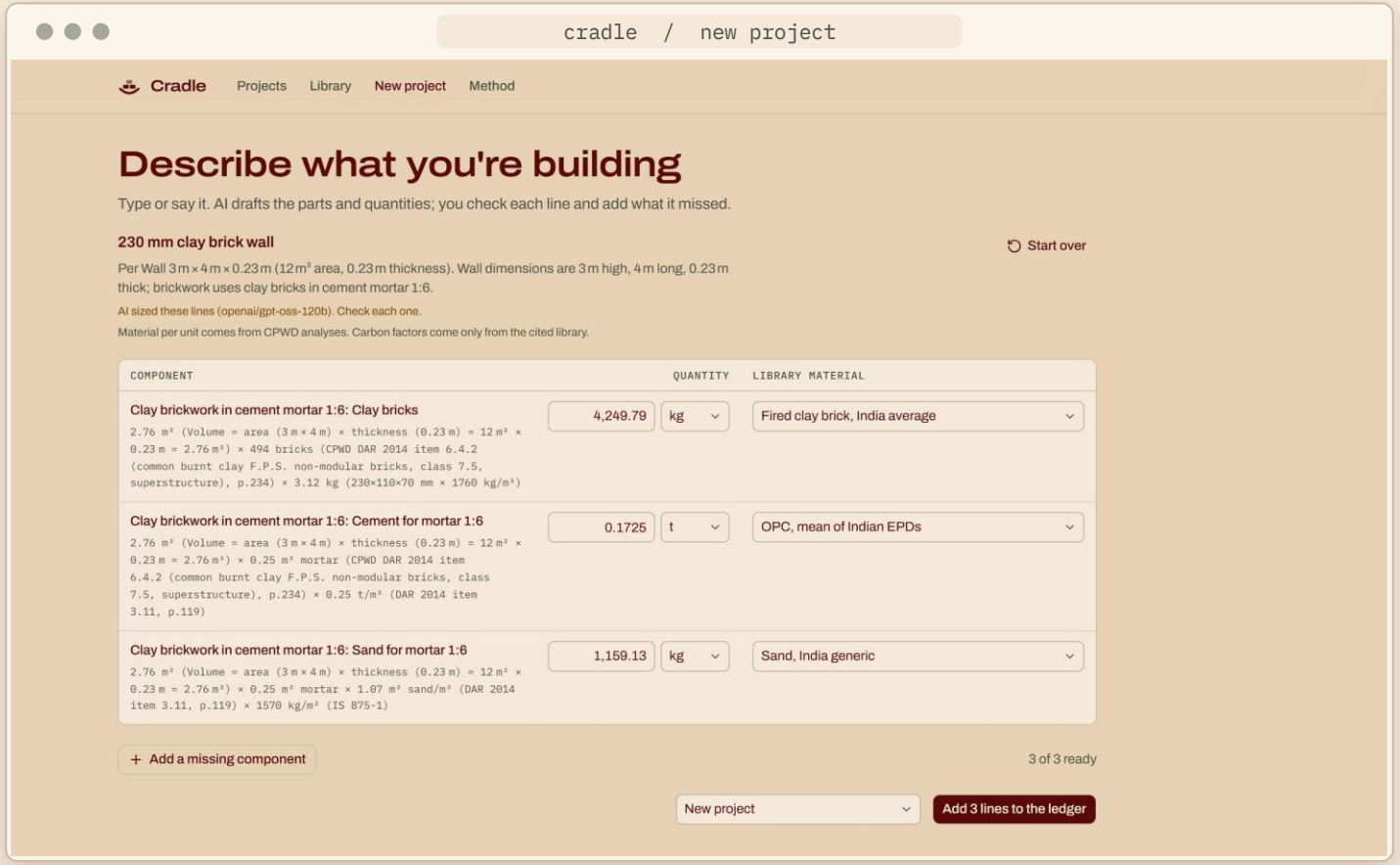
1. **You type or say**
   "230 mm clay brick wall, 3 m × 4 m"

2. **AI only sizes it**
   3 m × 4 m × 0.23 m = 2.76 m³

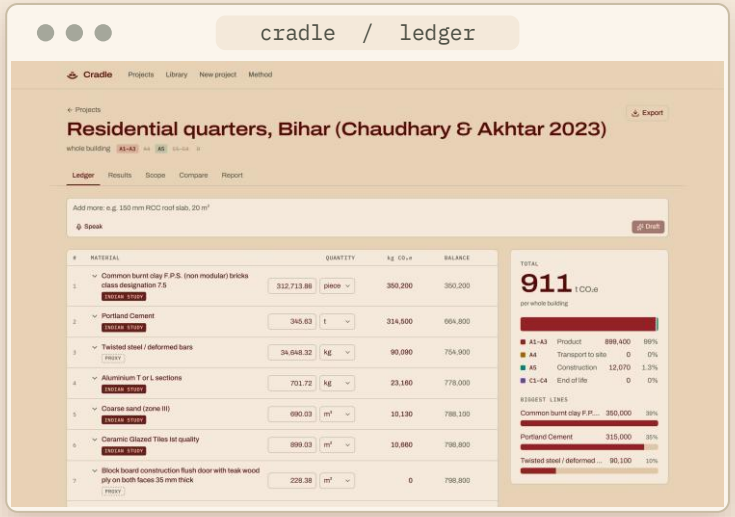
3. **CPWD gives the materials**
   DAR 2014 item 6.4.2: 494 bricks + 0.25 m³ mortar per m³

4. **Library gives the factors**
   You check each line, then add it to the ledger

The AI never picks a factor. When it did, bricks and mortar came out 2× wrong.

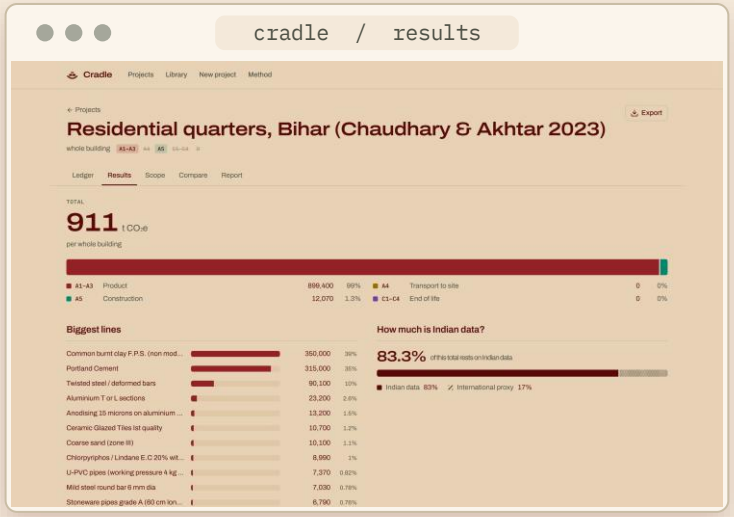


From the first line to the final report



01 Ledger

Every material with its quantity, data tag and running total



02 Results

Total by module, the biggest lines and the Indian-data share



03 Report

ISO 14044 order; exports Excel, BibTeX and the project file

REAL SCREENS OF CRADLE · THE VALIDATED BIHAR BUILDING, 78 LINES

Cradle reproduces a published Indian study

CHAUDHARY & AKHTAR (2023), ACS ENVIRONMENTAL AU · RESIDENTIAL QUARTERS, BHOJPUR, BIHAR · 78 MATERIALS FROM A CPWD BOQ

A1–A3 DIFFERENCE

0.00

kg CO_2e, on 899,412.37 kg

The paper's own quantities, units and factors were entered, so any gap could only come from the calculation itself.

78 of 78 lines match

49 automated tests pass

STAGE	PUBLISHED	CRADLE	DIFFERENCE
A1–A3 materials	899,412.37	899,412.37	**0.00**
A5 construction	12,075.01	12,074.60	0.41 (0.003%)
A4 transport	24,631	not compared	see note

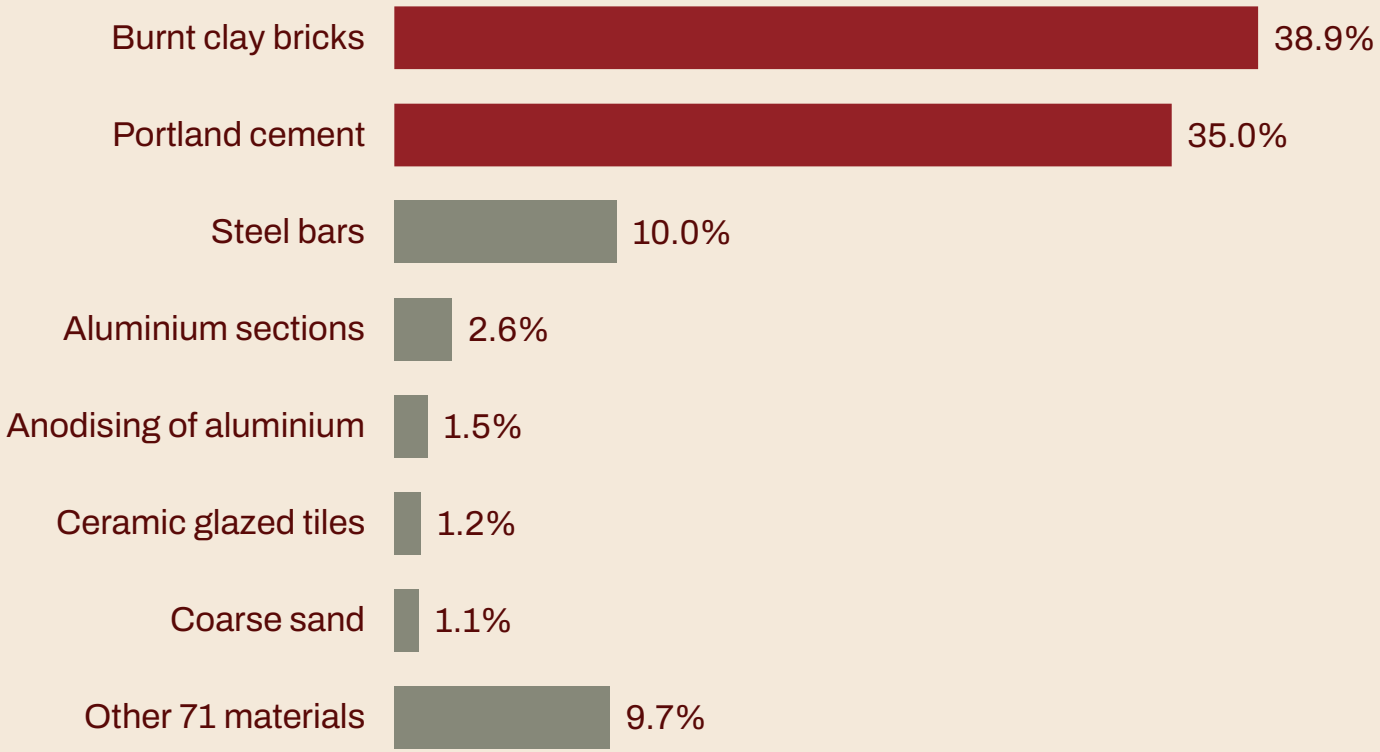
KG CO_2E

A5 The 0.41 kg gap is one welding item the paper gives per cm, a unit Cradle does not have.

A4 The paper rounds whole truck trips per group; Cradle shares trips by mass. A4 is 2.6% of the total.

Bricks and cement are three-quarters of the total

SHARE OF A1–A3 EMISSIONS · TOTAL 899.4 t CO_2e · 78 MATERIALS



74%

from bricks and cement alone, so their factors matter most

83.3%

of the total rests on Indian data; the rest on foreign factors the paper used

Fly ash brick and AAC walls carry about half

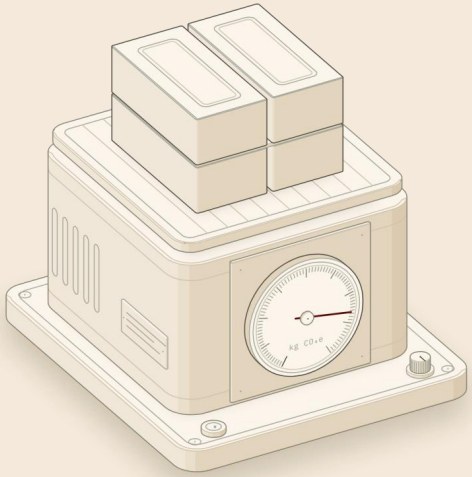


FIG 3a

Clay brick, mortar 1:6

665 kg CO_2e

BASELINE

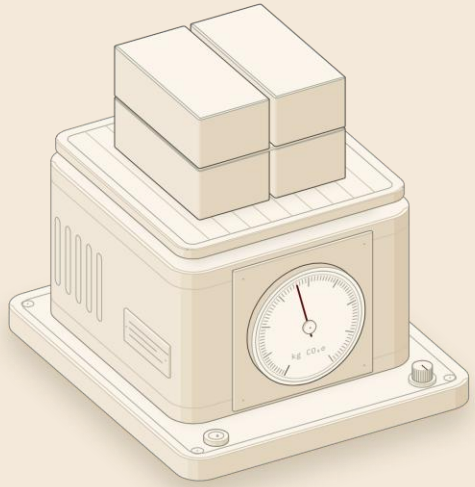


FIG 3b

Fly ash–lime brick, mortar 1:6

361 kg CO_2e

−46%

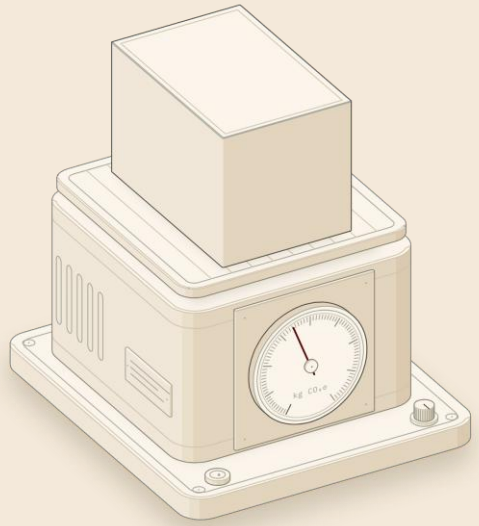


FIG 3c

AAC block, mortar 1:4

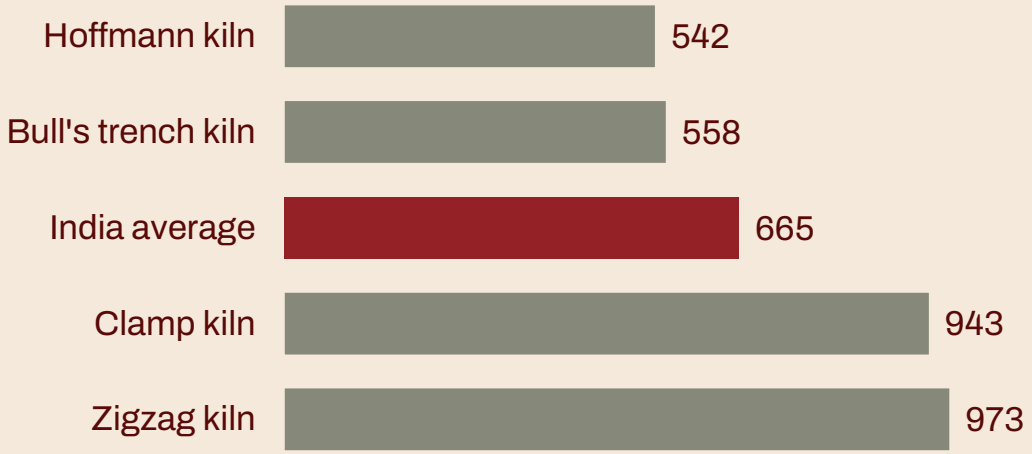
342 kg CO_2e

−49%

A1–A3 · QUANTITIES FROM CPWD DAR 2014 (ITEMS 6.4.2, 6.34.2, 6.38) · FACTORS FROM THE LIBRARY · THE SAME SCALE IS THE APP'S HOME FIGURE

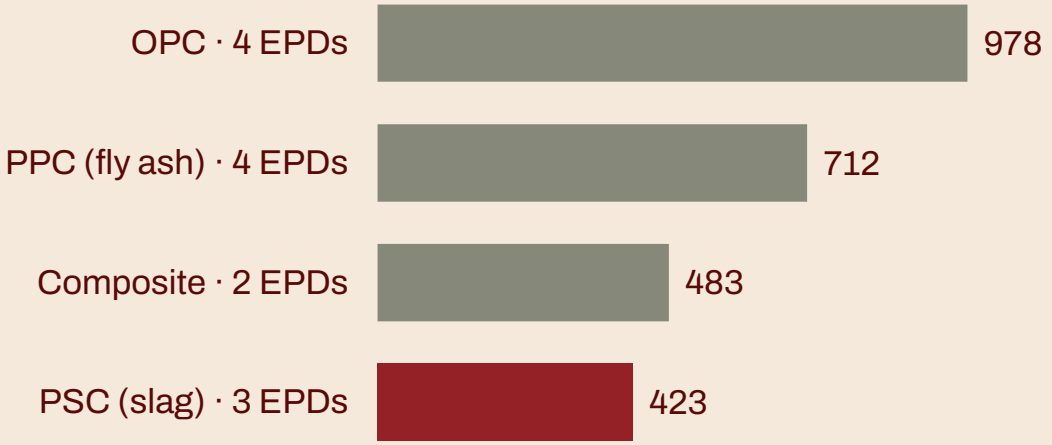
The factor you pick can matter more than the wall

SAME CLAY WALL, FIVE BRICK FACTORS · kg CO_2e PER m^3



1.8× from the kiln assumption alone: more than the gap between clay brick and AAC.

CEMENT TYPES, MEAN OF INDIAN EPDs · kg CO_2e PER TONNE



Under half of OPC for slag cement. Even OPC brands range 921 to 1,010.

A result should always say which factor it used, and where it came from.

What the project delivers

01 A free calculator for EN 15804 modules A1–A3, A4, A5, C1–C4 and D, with the working shown for every number.

02 106 Indian factors, each cited to its table or page; 54 come from Indian EPDs.

03 It reproduces a published Indian building: A1–A3 exactly, A5 within 0.41 kg.

04 Per m^3, fly ash brick (361) and AAC (342) walls are about half of clay brick (665 kg CO_2e).

05 The brick factor alone moves a clay wall from 542 to 973 kg CO_2e/m^3, so the factor must be stated.

NEXT STEPS

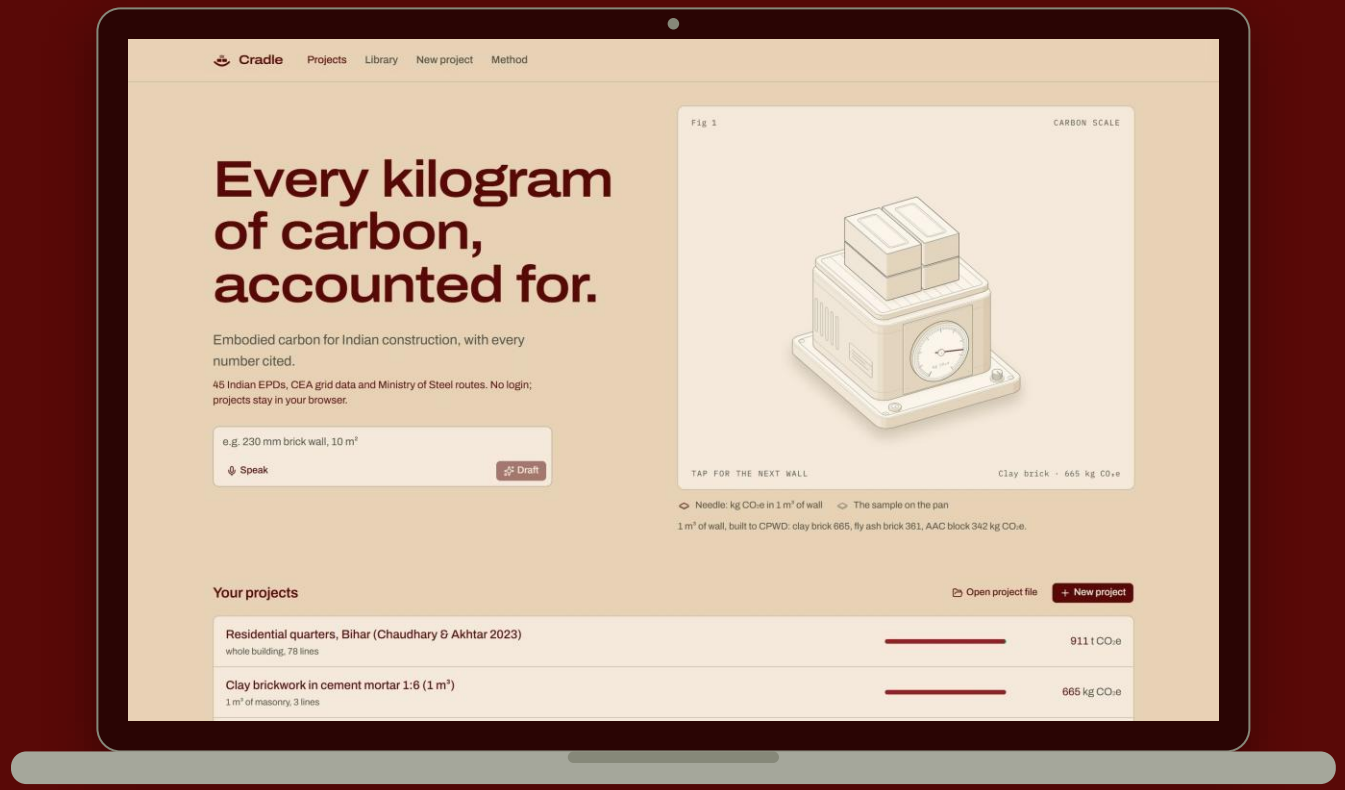
- A second validation case (Patil et al. 2023)
- A4 by whole truck trips, then re-check
- Published ranges on more factors (9 of 106)
- Compare walls per m^2 of equal function
- Indian end-of-life data for C1–C4 and D
- New Indian EPDs for bricks and blocks

Thank you

Questions are welcome.

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CRADLE · HOME